# Week 03: Computation and Mind: Turing Machines, Symbol Systems and Functionalism

## Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

## Overview

The claim that machines could think rests on a view of what thinking is. This week introduces the theory of computation, the physical symbol system hypothesis and functionalism, the doctrines that together make strong AI a coherent possibility [1, 2, 3].

## Learning outcomes

By the end of the week, students are expected to describe a Turing machine and the Church-Turing thesis. They should be able to state the physical symbol system hypothesis and the computational theory of mind, and to explain functionalism and multiple realisability. Finally, they should be able to implement and test simple Turing machines.

## Lecture

### What computation is

In 1936, Turing analysed what a human computer, a person carrying out a calculation by rule, can do, and captured it in an abstract machine [1]. A Turing machine has an unbounded tape divided into cells and a head that reads and writes one symbol at a time and moves left or right. It also has a finite set of states and a transition table. For each state and symbol, the table fixes what to write, where to move and which state comes next. Turing also described a universal machine that can simulate any other machine given its description, the idea behind the stored-program computer. Figure 3.1 shows a machine that adds one to a binary number.

In the same year, Church reached equivalent results with a different formalism [4]. The Church-Turing thesis states that every function that can be computed by an effective procedure can be computed by a Turing machine. It is a thesis rather than a theorem, because the notion of an effective procedure is informal, but every alternative formalism proposed since has turned out to be equivalent.

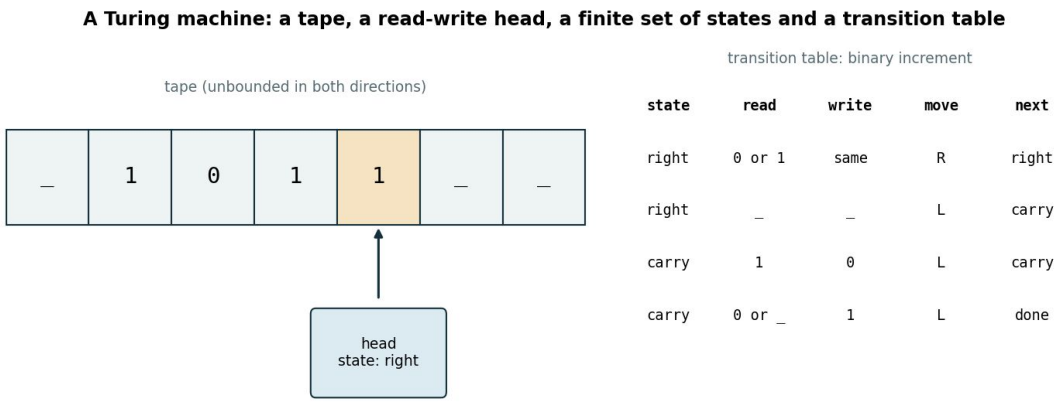



*Figure 3.1. A Turing machine for adding one to a binary number: The head moves to the right end, then propagates the carry to the left [1].*

## Symbols, search and the language of thought

Newell and Simon turned computation into a hypothesis about intelligence [2]. Their physical symbol system hypothesis states that a physical symbol system has the necessary and sufficient means for general intelligent action. A physical symbol system is a machine that creates, copies, modifies and destroys symbol structures, and its symbols designate objects and processes. If the hypothesis is true, human intelligence is a form of symbol processing, and a suitably programmed computer can in principle be intelligent.

Fodor developed a related view in philosophy of mind, the computational theory of mind [5]. Thinking consists of computations over mental representations that have a language-like structure, a language of thought. The view explains how thought can be productive, since new thoughts are built from old parts, and it links the meaning of thoughts to their causal roles in reasoning.

## Functionalism and multiple realisability

Putnam compared mental states with the logical states of a Turing machine [3]. A machine state is defined by its relations to inputs, outputs and other states, not by the hardware that realises it. Functionalism generalises the idea: A mental state such as pain is defined by its causal role, what causes it, what it causes and how it interacts with other states. It follows that the same mental state could be realised in very different substrates, such as neurons or silicon. This is multiple realisability, the premise that makes strong AI coherent. If minds are defined by function, the right program on any capable hardware would have a mind.

The interactive lab makes multiple realisability concrete. The same function, adding one to a number, is realised by a Turing machine, by a circuit of logic gates and by a lookup table. The three realisations agree on every input and differ completely in their internal operations.

### When does a physical system compute?

Functionalism faces a puzzle. If computing is merely a matter of mapping physical states onto computational states, almost any physical system could be said to compute almost anything, which would make computationalism trivially true. Piccinini proposed a mechanistic account in response [6]. On this account, a physical system computes when it is a mechanism whose function is to process vehicles, such as digits. The processing follows rules that are sensitive to properties of those vehicles. The account separates genuine computers from rocks and walls and gives computationalism empirical content.

The doctrines of this week set up the next one. If thinking is computation and computation is substrate-independent, then running the right program suffices for a mind. Searle's Chinese Room argument, the topic of Week 4, targets exactly this inference.

> ***Pause and reflect.*** *Is a thermostat a physical symbol system? Apply Newell and Simon's definition and Piccinini's criterion and compare the answers.*

## Hands-on components

Interactive lab: Part A simulates four Turing machines step by step, including a palindrome checker and the two-state busy beaver. Part B realises the same function with a Turing machine, a logic circuit and a lookup table to illustrate multiple realisability [1, 3]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-03/lab.html

Colab notebook: It implements a Turing machine simulator and verifies that the binary increment machine realises the same function as ordinary arithmetic. It then runs the two-state busy beaver, and two exercises analyse small machines [1, 3]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-03/NB03_computation_and_mind.ipynb

## Reflection prompts

1. If functionalism is true, what would it take for a program to have your beliefs? What would be missing if it had only your input-output behaviour?
2. Which of the three realisations in the interactive lab, if any, seems most like thinking? Why do intuitions differ between them?
3. Is the brain a physical symbol system? Give one consideration for and one against.

## Weekly task and submission

**What to do.** Explain how three ideas of this week combine into an argument for strong AI. Write the argument as numbered steps that use the Church-Turing thesis, the physical symbol system hypothesis and functionalism. Then identify the weakest link in the chain and defend your choice.

**Where to look.** Lecture: sections "What computation is", "Symbols, search and the language of thought" and "Functionalism and multiple realisability". The section "When does a physical system compute?" raises a problem for one of the links. Interactive lab: Part B, where one function is realised in three ways. Colab notebook: Section 2, which checks that a Turing machine and Python's arithmetic realise the same function.

**Example.** Step 1 could read: Every effective procedure can be carried out by a Turing machine. Step 2 could read: Thinking is the manipulation of symbols by effective procedures. A student who doubts Step 2 can argue that it is the weakest link, because the Church-Turing thesis says nothing about minds.

**What to submit.** About 400 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

## References

[1] Turing, A. M. (1936). On computable numbers, with an application to the Entscheidungsproblem. Proceedings of the London Mathematical Society, s2-42(1), 230-265. https://doi.org/10.1112/plms/s2-42.1.230

[2] Newell, A., & Simon, H. A. (1976). Computer science as empirical inquiry: Symbols and search. Communications of the ACM, 19(3), 113-126. https://doi.org/10.1145/360018.360022

[3] Putnam, H. (1960). Minds and machines. In S. Hook (Eds.), Dimensions of Mind (pp. 138-164). New York University Press.

[4] Church, A. (1936). An unsolvable problem of elementary number theory. American Journal of Mathematics, 58(2), 345-363. https://doi.org/10.2307/2371045

[5] Fodor, J. A. (1975). The Language of Thought. Thomas Y. Crowell.

[6] Piccinini, G. (2015). Physical Computation: A Mechanistic Account. Oxford University Press.